

HDOC Day-22

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SUBJECT: C programming

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Problem Statement

Write a menu-driven C program that takes two positive integers, m and n, with $m \leq n$. The program should keep showing the menu until the user chooses Exit. A switch statement is used to perform the selected operation.

1. Print the multiplication tables of m and n side by side, using multipliers 1 to 10.
2. Display a multiplication grid containing products of integers 1 to m and 1 to n.
3. For each integer from m to n, print that integer followed by the next five consecutive integers.
4. Exit the program. If the user enters any other menu choice, show a simple error message and display the menu again.

Algorithm

1. Start the program.
2. Read two positive integers m and n.
3. Check the input. If either value is not positive or $m > n$, print an error message and stop.
4. Show the menu and ask the user to enter a choice.
5. Use switch to perform the selected task: print both tables, display the multiplication grid, print consecutive integers, or exit.
6. After completing an option, show the menu again. Keep repeating until the user selects 4.
7. End the program.

Test-Case Table

Test Case	m, n	Choice	Expected Result	Status
TC1 - Multiplication tables	3 5	1	Prints tables of 3 and 5 side by side from 1 to 10.	Pass
TC2 - Multiplication grid	3 5	2	Displays a 3 x 5 multiplication grid.	Pass
TC3 - Consecutive integers	3 5	3	For 3, 4 and 5, prints each number followed by the next five integers.	Pass

TC4 - Invalid menu choice	3 5	7	Displays an invalid menu choice message and shows the menu again.	Pass
TC5 - Exit	3 5	4	Displays the exit message and terminates the program.	Pass
TC6 - Invalid m and n	6 3	-	Rejects the input because $m > n$.	Pass

C Program

```
#include <stdio.h>
```

```
int main()
{
    int m, n, choice, i, j;

    printf("Enter two positive integers m and n (m <= n): ");
    scanf("%d %d", &m, &n);

    if (m <= 0 || n <= 0 || m > n)
    {
        printf("Invalid input. Enter positive integers with m <= n.\n");
        return 0;
    }

    do
    {
        printf("\n1. Multiplication tables of m and n\n");
        printf("2. Multiplication grid\n");
        printf("3. Consecutive integers\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("\nMultiplication Tables of %d and %d\n", m, n);
                for (i = 1; i <= 10; i++)
                    printf("%d x %d = %d\t\t%d x %d = %d\n",
                        m, i, m * i, n, i, n * i);
                break;

            case 2:
                printf("\nMultiplication Grid (m=%d, n=%d)\n\n", m, n);
                for (j = 1; j <= n; j++)
                    printf("%4d", j);
                printf("\n");

                for (i = 1; i <= m; i++)
                {
                    printf("%2d |", i);
                    for (j = 1; j <= n; j++)
                        printf("%4d", i * j);
                    printf("\n");
                }
                break;

            case 3:
                printf("\nConsecutive Integers from %d to %d\n", m, n);
                for (i = m; i <= n; i++)
                {
                    printf("%d: ", i);
                    for (j = i; j <= i + 5; j++)
                        printf("%d ", j);
                    printf("\n");
                }
                break;
        }
    }
}
```

```
        case 4:
            printf("Exiting program.\n");
            break;

        default:
            printf("Invalid menu choice. Please enter 1 to 4.\n");
    }
} while (choice != 4);

return 0;
}
```

Compilation and Execution

I saved the program as day22.c. In the Kali Linux terminal, it can be compiled and run using:

```
gcc day22.c
```

```
./a.out
```

The screenshots below show the program being checked with the prepared test cases.

TC1 - Multiplication tables

Input: m, n = 3 5 | Menu choice = 1

Expected: Prints tables of 3 and 5 side by side from 1 to 10.

```
(kali@kali) - [~/Desktop/cprog]
$ ./a.out
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 3 and 5
3 x 1 = 3      5 x 1 = 5
3 x 2 = 6      5 x 2 = 10
3 x 3 = 9      5 x 3 = 15
3 x 4 = 12     5 x 4 = 20
3 x 5 = 15     5 x 5 = 25
3 x 6 = 18     5 x 6 = 30
3 x 7 = 21     5 x 7 = 35
3 x 8 = 24     5 x 8 = 40
3 x 9 = 27     5 x 9 = 45
3 x 10 = 30    5 x 10 = 50
```

Result: PASS

TC2 - Multiplication grid

Input: m, n = 3 5 | Menu choice = 2

Expected: Displays a 3 x 5 multiplication grid.

```
(kali@kali) - [~/Desktop/cprog]
$ ./a.out
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

Multiplication Grid (m=3, n=5)

      1  2  3  4  5
1 |  1  2  3  4  5
2 |  2  4  6  8 10
3 |  3  6  9 12 15
```

Result: PASS

TC3 - Consecutive integers

Input: m, n = 3 5 | Menu choice = 3

Expected: For 3, 4 and 5, prints each number followed by the next five integers.

```
(kali@kali) - [~/Desktop/cprog]
$ ./a.out
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 3

Consecutive Integers from 3 to 5
3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10
```

Result: PASS

TC4 - Invalid menu choice

Input: m, n = 3 5 | Menu choice = 7

Expected: Displays an invalid menu choice message and shows the menu again.

```
(kali@kali) - [~/Desktop/cprog]
$ ./a.out
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 7
Invalid menu choice. Please enter 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
```

Result: PASS

TC5 - Exit

Input: m, n = 3 5 | Menu choice = 4

Expected: Displays the exit message and terminates the program.

```
(kali@kali) - [~/Desktop/cprog]
$ ./a.out
Enter two positive integers m and n (m <= n): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4
Exiting program.
```

Result: PASS

TC6 - Invalid m and n

Input: m, n = 6 3

Expected: Rejects the input because $m > n$.

```
(kali@kali) - [~/Desktop/cprog]
$ ./a.out
Enter two positive integers m and n (m <= n): 6 3
Invalid input. Enter positive integers with m <= n.
```

Result: PASS

Conclusion

The program worked correctly for all the prepared test cases. It handled all three operations, an invalid menu choice, the Exit option, and invalid values of m and n.